

German Electricity Market Analysis

Bronze → Silver → Gold pipeline on real Bundesnetzagentur / SMARD data, Jan 2024 – Dec 2025

Tools

Python • Pandas • SMARD REST API (smard.de/app/chart_data) • matplotlib — pipeline structured to map onto a GCP BigQuery + dbt Medallion setup (Bronze/Silver/Gold), same architecture as the German Motorcycle Market Analysis project.

Data sources

- SMARD.de — Bundesnetzagentur's official electricity market data platform (CC BY 4.0). Monthly generation (MWh) for six sources, total system load, and the average DE/LU day-ahead auction price, January 2024 through December 2025 (24 months).

Key findings

- **Solar had the strongest year of any source:** +16.8% generation growth in 2025 vs 2024, far ahead of natural gas (+6.4%) and hard coal (+3.0%).
- **Lignite (-5.4%) and onshore wind (-5.2%) both declined** — coal continuing its structural phase-down, while onshore wind simply had a weaker wind year.
- **Renewable share swung from 39.7% to 74.4%** month to month — June 2025 was the greenest month in the dataset (74.4%), February 2025 the least green (39.7%), a wider spread than the same months in 2024.
- **The 'Dunkelflaute' effect shows up clearly in price data:** February 2025's renewable dip (39.7%) coincided with the highest average day-ahead price in the two-year window (€128.52/MWh) — low wind-and-solar months are consistently the most expensive months, in both 2024 and 2025.

Charts

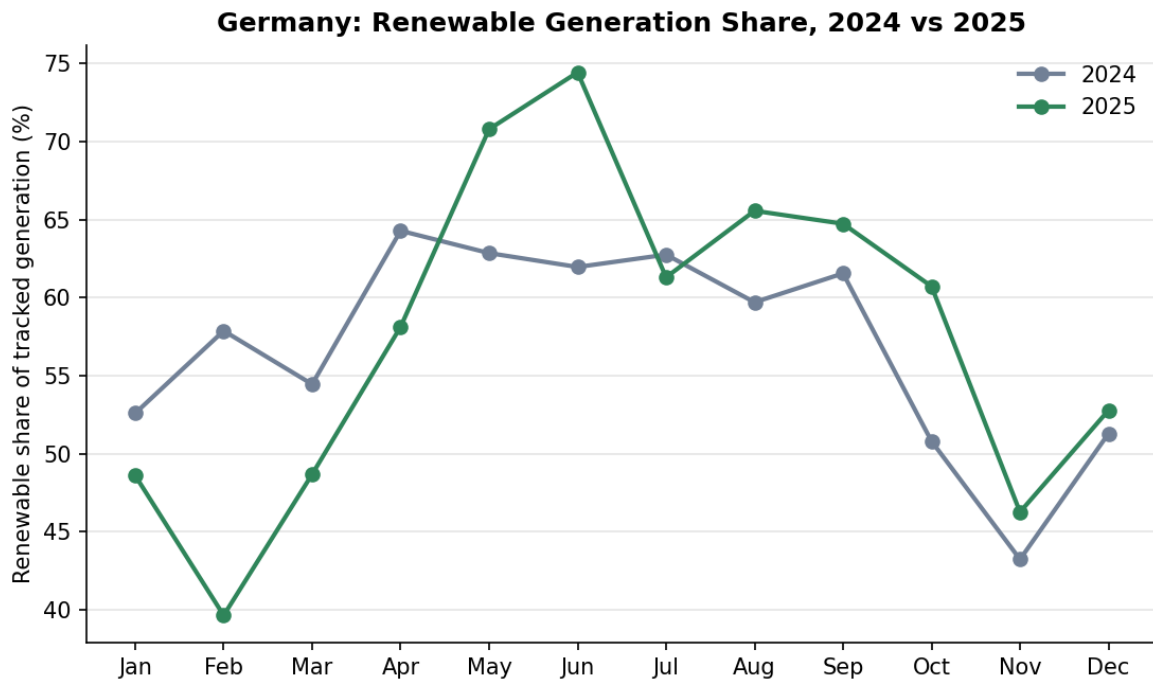


Figure 1. Renewable generation share, 2024 vs 2025, by calendar month.

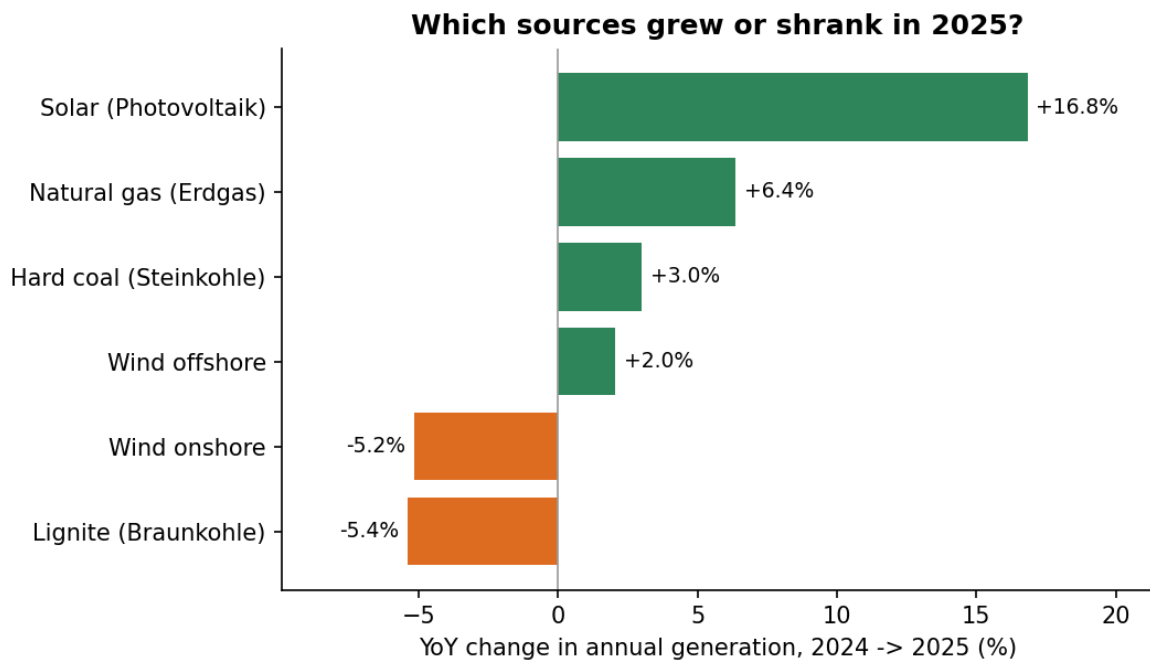


Figure 2. Year-over-year change in annual generation by source, 2024 to 2025.

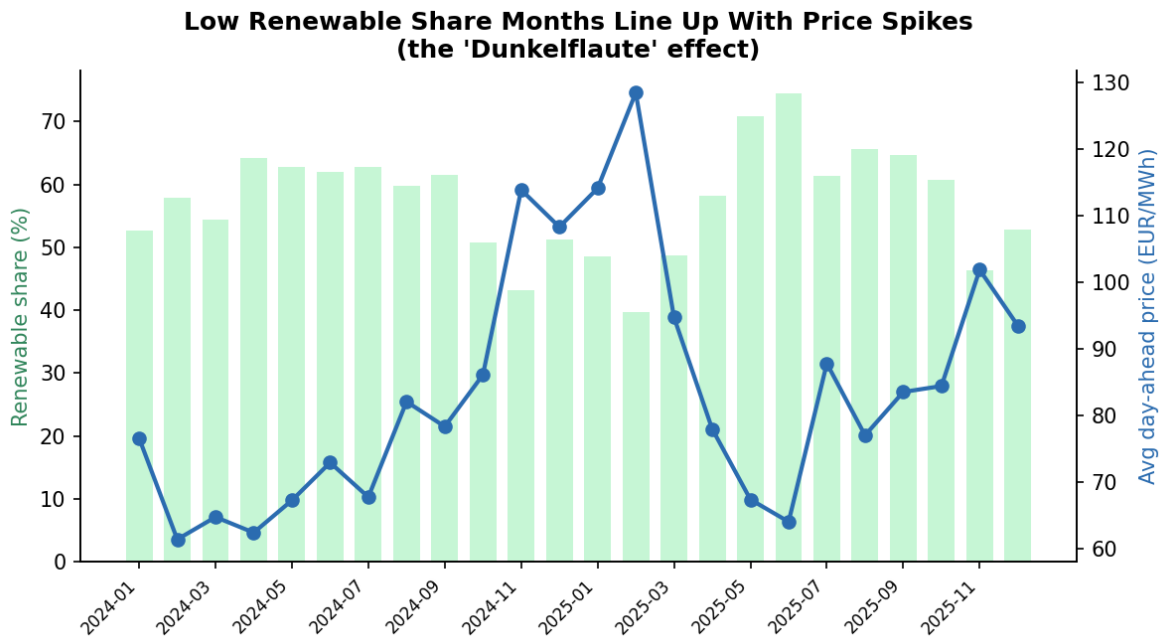


Figure 3. Average day-ahead price vs renewable share by month — the Dunkelflaute effect.

Notes on data

Generation totals cover the six SMARD filters with the most complete 2024-2025 history (lignite, hard coal, natural gas, onshore wind, offshore wind, solar); smaller categories (biomass, hydro, pumped storage, "other") are not included, so renewable/fossil shares are relative to this tracked subset, not all of German generation. All figures pulled directly from the SMARD API on 2026-09-14 and not modified beyond unit conversion (MWh → GWh) and timezone localization.